EDS 223 HW 2: Exploring patterns of environmental justice

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```
library(tidyverse)
library(tmap)
library(sf)
library(here)
library(janitor)
library(knitr)
```

```
# Read in the data
ejscreen <- st_read(here("data", "ejscreen","EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI
    clean_names()
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/Users/emaliapartlow/Desktop/MEDS/eds223/eds223-
hw2/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```
holc <- st_read(here("data", "mapping-inequality","mapping-inequality-los-angeles.json"))
    clean_names() |>
    st_transform(crs = st_crs(ejscreen)) # Transform the CRS to match ejscreen
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `/Users/emaliapartlow/Desktop/MEDS/eds223/eds223-hw2/data/mapping-inequality/mapping-
inequality-los-angeles.json'
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```
crashes <- read_csv(here("data", "SWITRS_LA","crashes.csv")) |>
           clean_names() |>
           as_tibble() |>
           drop_na(c('point_x', 'point_y')) |> # Remove NAs from x and y columns
           st_as_sf(coords = c("point_x", "point_y"), crs = 4326) |> # Create a geometry
           st_transform(crs = st_crs(ejscreen)) # Transform back to a CRS that matches ej
```

```
# Double check all dataframe's CRS match
if((st_crs(ejscreen) == st_crs(holc)) & (st_crs(ejscreen) == st_crs(crashes))){
  print("It's a match")
} else {
  warning("Coordinate refrence systems do not match") # Include a warning if the datafram
}
```

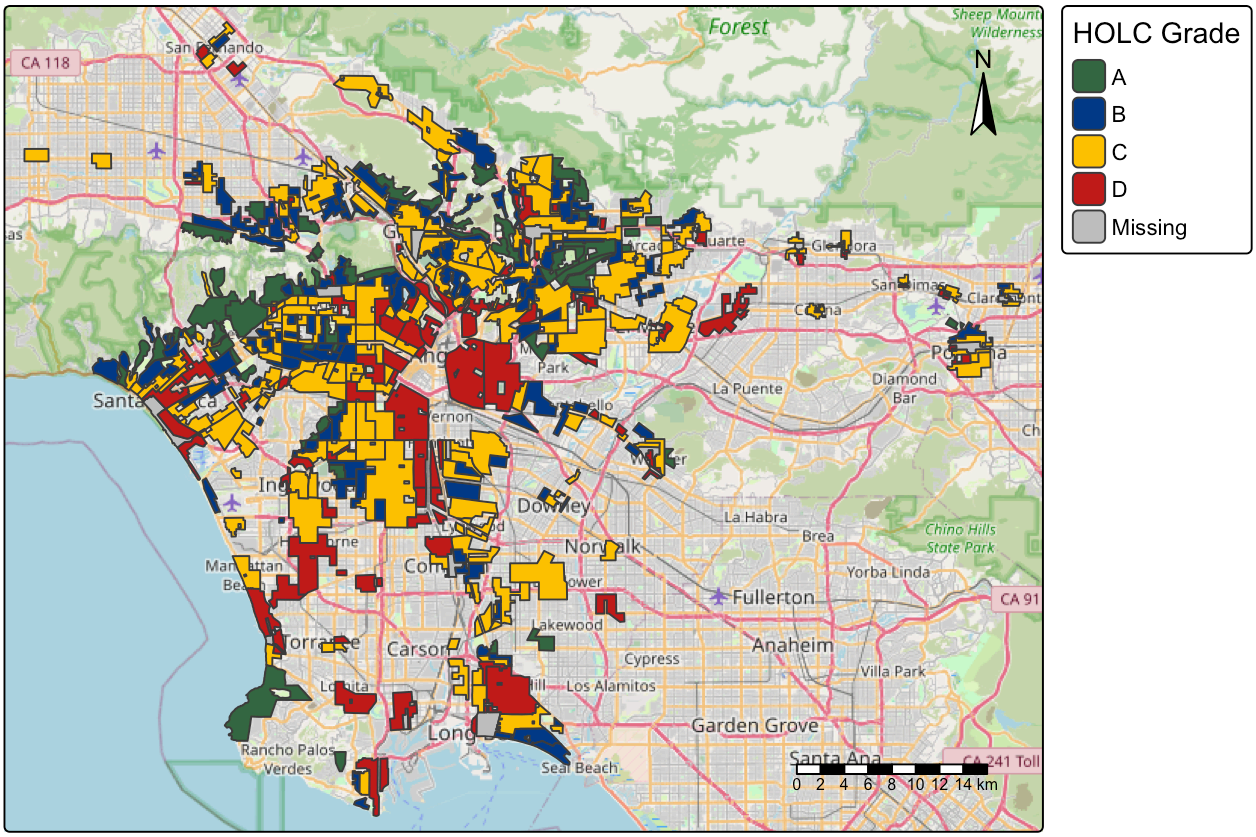
```
[1] "It's a match"
```

Part 1: Legacy of redlining in current environmental (in)justice

```
# Create a color palette for the first map's holc grade
holc_grade <- c('A' = '#386641', 'B' = '#023e8a', 'C' = '#ffc300', 'D' = '#c1121f')
```

```
# Create a map of historical redlining neighborhoods
tm_shape(holc) +
    tm_polygons(
        fill = 'grade', # Fill polygons based on holc grade
        fill.scale = tm_scale(values = holc_grade), # Use palette created in previous ste
        fill.legend = tm_legend(title = 'HOLC Grade',
                                position = tm_pos_out('right', 'center'))
    ) +
tm_compass(position = c('right', 'top')) +
tm_scalebar(position = c('right', 'bottom')) +
tm_title('Historical redlining neighborhoods in LA') +
tm_legend(position = tm_pos_out('right', 'center')) +
tm_basemap('OpenStreetMap')
```

Historical redlining neighborhoods in LA



The map displays neighborhoods and their assigned HOLC grades, from A (highest perceived safety for real estate investment in green) to D (most hazardous in red).

```
# Join holc and ejscreen dataframes spatially
ej_holc <- st_join(ejscreen, holc) |> # Left join, st_intersects
  filter((cnty_name == 'Los Angeles County')) # Filter for LA county
```

```
# Create a summary dataframe of the percentage of census block groups that fall within ea
summary_census <- ej_holc |>
  st_drop_geometry() |> # Drop the geometry column
  group_by(grade) |> # Group by holc grade
  summarise(total_grade = n()) |> # Creates column 'total_grade' with total counts for ea
  mutate('Percent of census block groups within grade' = (total_grade/sum(total_grade)) *
  rename( # Rename columns
    'HOLC grade' = 'grade',
    'Total count of census block groups within grade' = 'total_grade'
  )
```

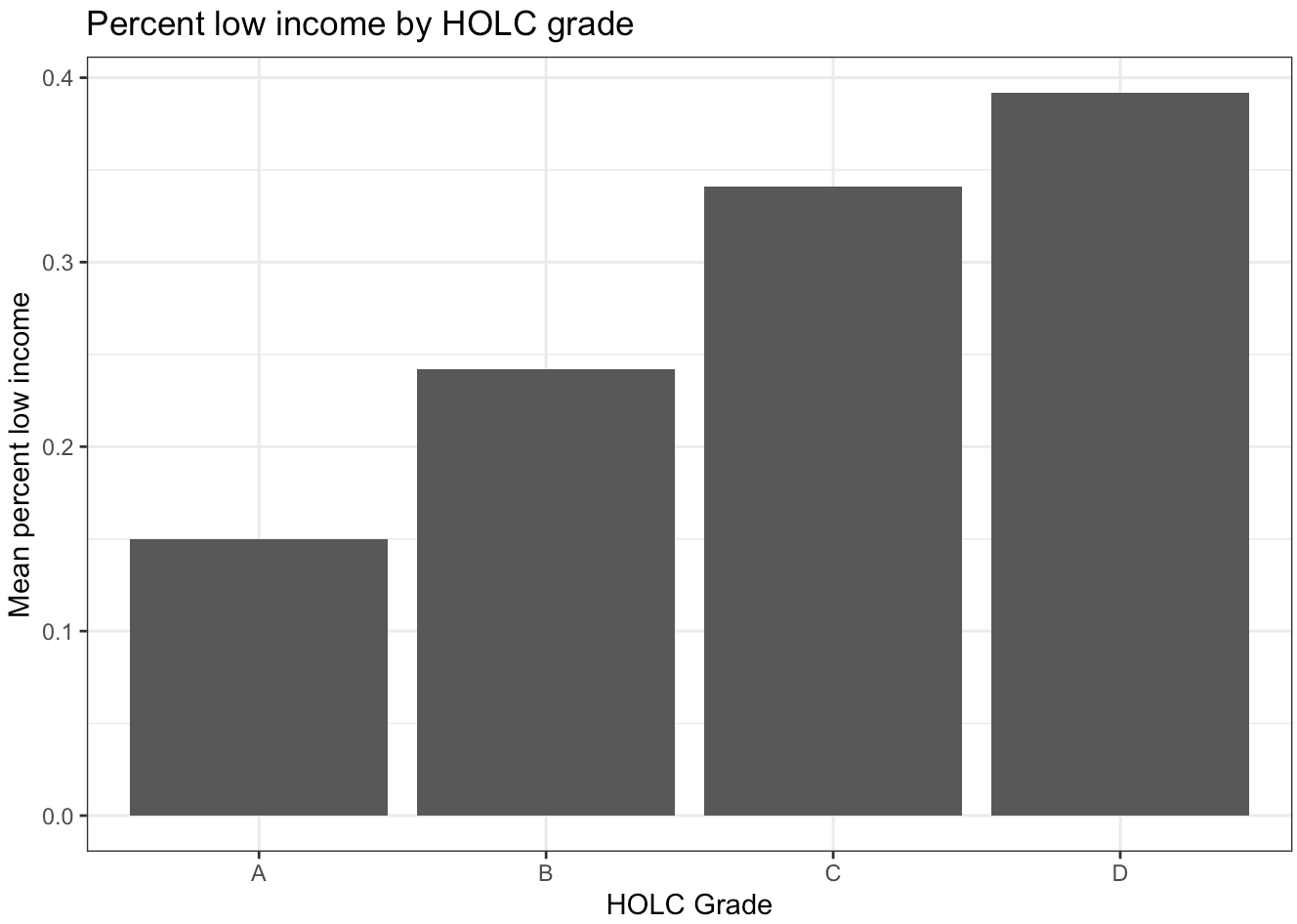
```
# Create a table of the summary_census data
kable(summary_census)
```

Table 1: Percentage of census block groups that fall within each HOLC grade

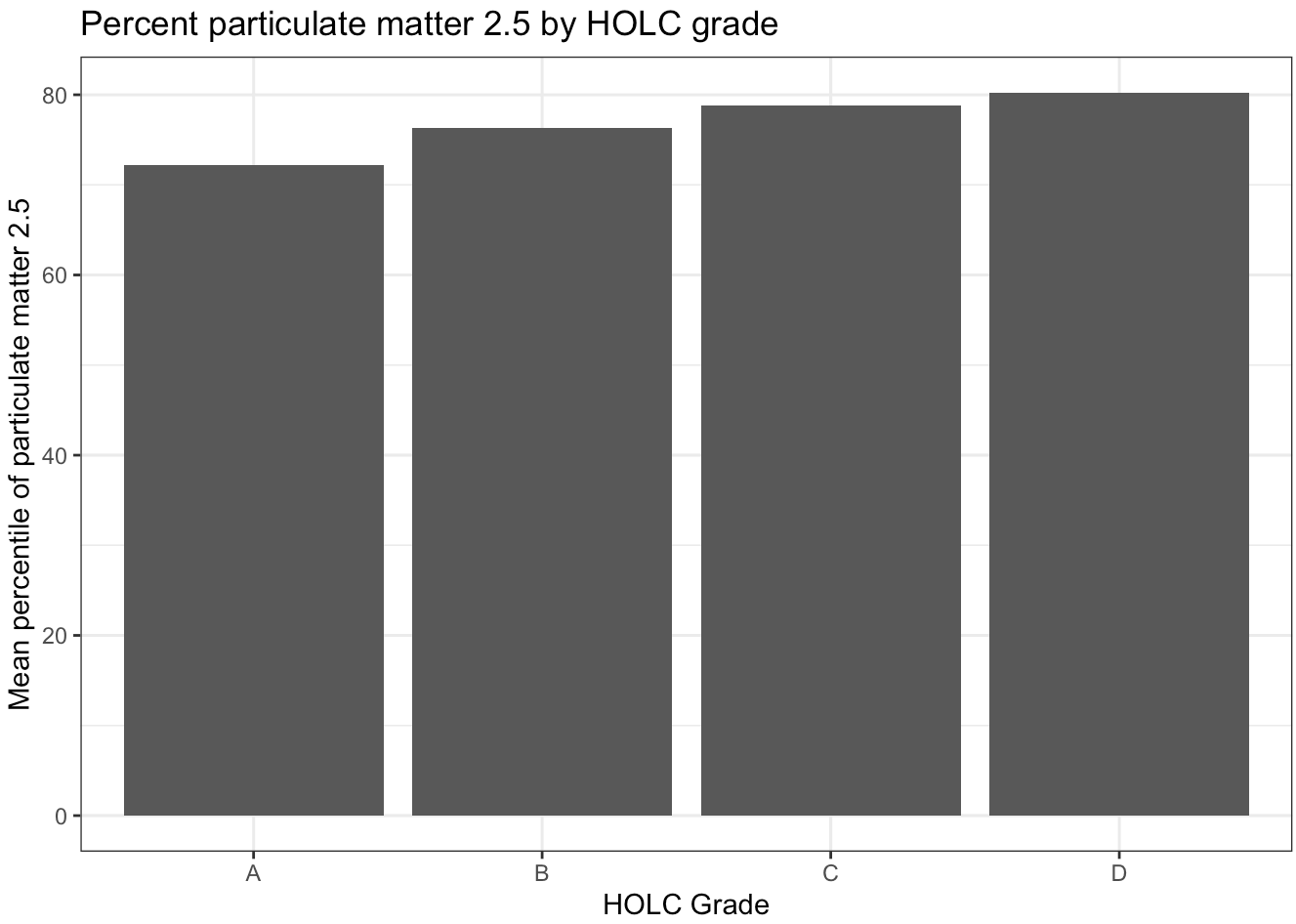
HOLC grade	Total count of census block groups within grade	Percent of census block groups within grade
A	449	4.99555
B	1239	13.78505
C	3058	34.02314
D	1346	14.97552
NA	2896	32.22074

```
# Create summary dataframe with means of 3 variables
three_means <- ej_holc |>
  st_drop_geometry() |> # Drop the geometry column
  group_by(grade) |> # Group by holc grade
  drop_na(grade) |> # Drop NAs from grade column
  summarise(
    mean_lowincpct = mean(lowincpct, na.rm = TRUE), # Mean percent low income
    mean_p_pm25 = mean(p_pm25, na.rm = TRUE), # Mean percentile for Particulate Matter 2.
    mean_lifeexppct = mean(lifeexppct, na.rm = TRUE) # Mean low life expectancy percent
    )
```

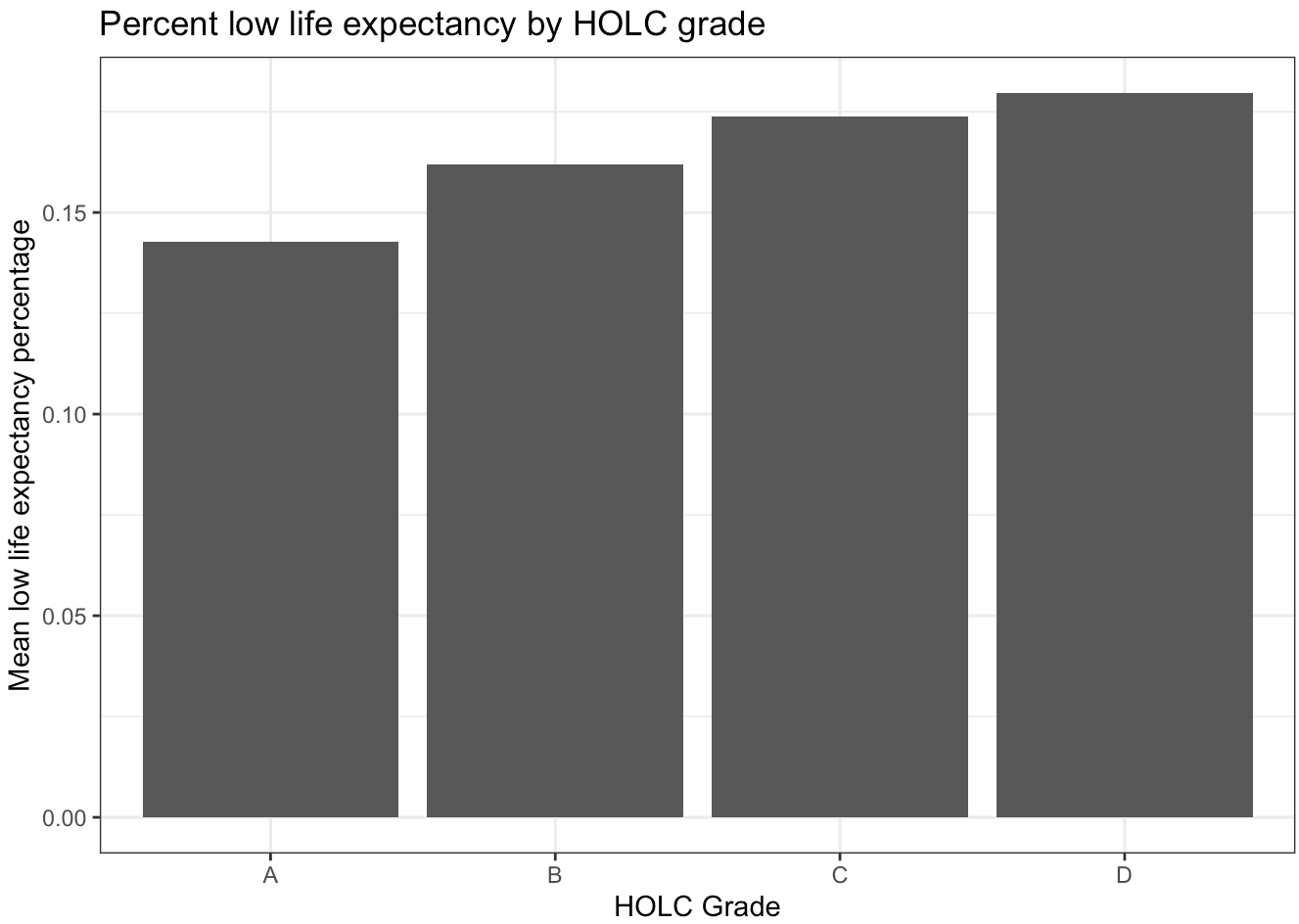
```
# Visualize low income
ggplot(
  data = three_means,
  mapping = aes(
    x = grade,
    y = mean_lowincpct
  )
) +
geom_col() +
theme_bw() +
labs(x = 'HOLC Grade', y = 'Mean percent low income', title = 'Percent low income by HOLC
```




```
# Visualize PM 2.5
ggplot(
  data = three_means,
  mapping = aes(
    x = grade,
    y = mean_p_pm25
  )
) +
geom_col() +
theme_bw() +
labs(x = 'HOLC Grade', y = 'Mean percentile of particulate matter 2.5', title = 'Percent
```




```
# Visualize low life expectancy
ggplot(
  data = three_means,
  mapping = aes(
    x = grade,
    y = mean_lifeexppct
  )
) +
geom_col() +
theme_bw() +
labs(x = 'HOLC Grade', y = 'Mean low life expectancy percentage', title = 'Percent low li
```



Percentages of low income people, low life expectancy, and particulate matter 2.5 are the lowest in areas with a HOLC grade of A, meaning they were ranked as having the best perceived safety for real estate investment. Those three variables are highest in HOLC grade D locations, that were considered hazardous. This suggests that a history of redlining in an area has socioeconomic and environmental consequences that continue to negatively impact the the people that live there.

Part 2: Legacy of redlining in pedstrian and bike accidents

```
# Join holc and crashes dataframes spatially
crashes_holc <- st_join(crashes, holc) |> # Left join, st_intersects
  select(grade, pedestrian_accident, bicycle_accident) |>
  filter(pedestrian_accident == 'Y' | bicycle_accident == 'Y') # Filter for rows where th
```

```
# Create a summary dataframe of the percentage of pedestrian and bicycle accidents that f
summary_crashes <- crashes_holc |>
  st_drop_geometry() |> # Drop geometry column
  group_by(grade) |> # Group by holc grade
  summarise(total_grade = n()) |> # Creates column 'total_grade' with total counts for ea
  mutate('Percent of pedestrian and bicycle accidents' = (total_grade/sum(total_grade)) *
  rename(
    'HOLC grade' = 'grade',
```

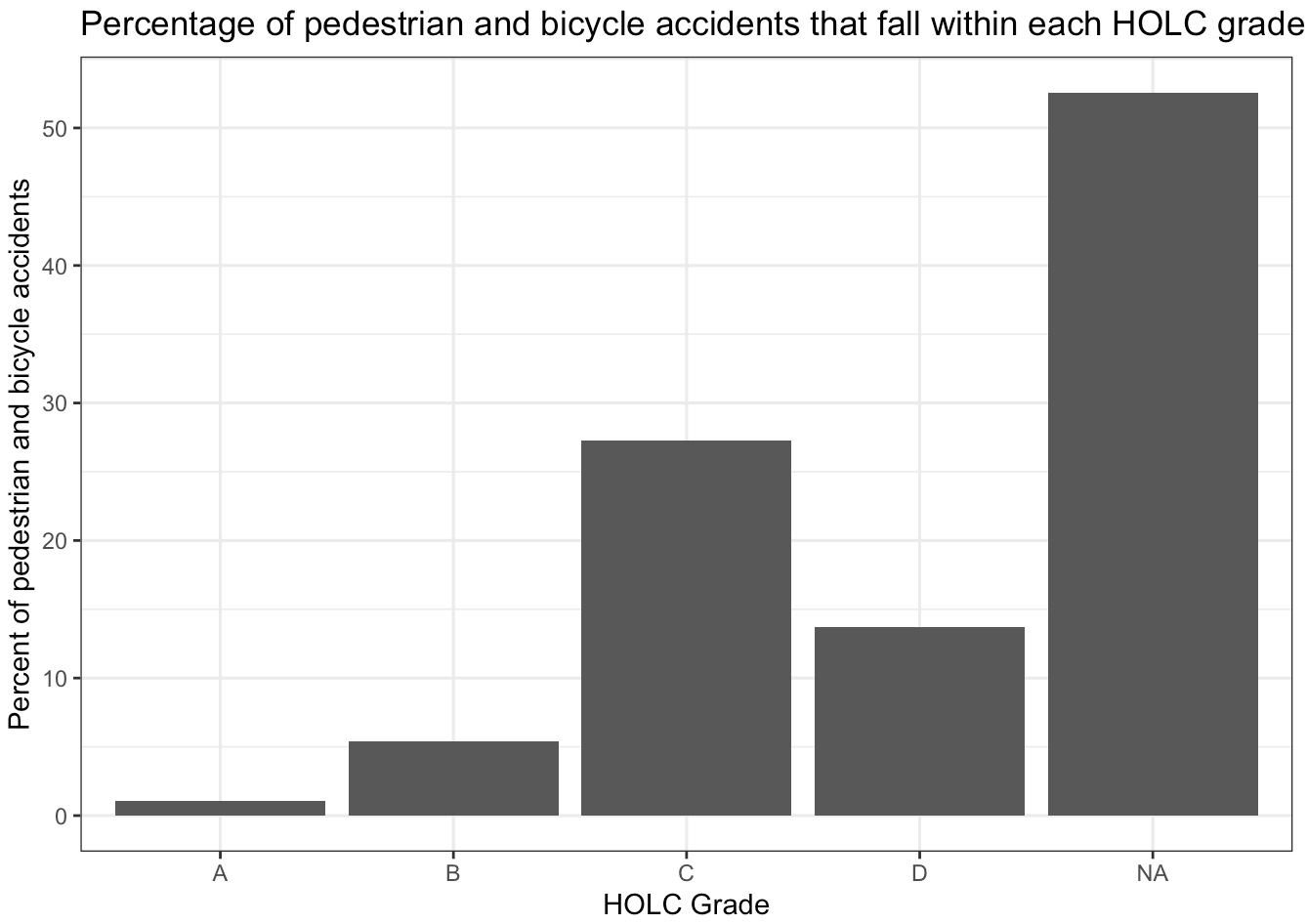
```
    'Total count of pedestrian and bicycle accidents' = 'total_grade'
  )
```

```
# Create a table of the summary_crashes data
kable(summary_crashes)
```

Table 2: Percentage of pedestrian and bicycle accidents that fall within each HOLC grade

HOLC grade	Total count of pedestrian and bicycle accidents	Percent of pedestrian and bicycle accidents
A	360	1.087646
B	1782	5.383849
C	9026	27.269706
D	4532	13.692257
NA	17399	52.566543

```
# Visualize crashes
ggplot(
  data = summary_crashes,
  mapping = aes(
    x = `HOLC grade`,
    y = `Percent of pedestrian and bicycle accidents`
  )
) +
geom_col() +
theme_bw() +
labs(x = 'HOLC Grade', y = 'Percent of pedestrian and bicycle accidents', title = 'Percen
```



Percentage of pedestrian and bicycle accidents are highest in grade C, with a vlaue of 27.27%, second highest in grade D, with a vlaue of 13.69%, and lowest in grade A, with a value of 1.09%. This suggests that pedestrians and bikers may be more vulnerable to accidents when they are in areas with a history of redlining.